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RESEARCH ARTICLE

Comparative Evaluation of GeneXpert MTB/RIF Ultra and GeneXpert MTB/RIF for Detecting Tuberculosis and Identifying Rifampicin Resistance in Pars Plana Vitrectomy Samples of Patients with Ocular Tuberculosis

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ABSTRACT

Background: Xpert MTB/RIF Ultra (Ultra) was evaluated for the first time on Ocular tuberculosis (OTB) samples and compared with Xpert.

Methods: Seventy five vitreous fluid samples (3 confirmed OTB, 47 clinically suspected OTB, and 25 controls) were subjected to Ultra, Xpert and Multiplex-PCR and compared against culture, composite reference standard (CRS), and gene sequencing.

Results: The sensitivity of Ultra was 50% in diagnosing OTB (100% against culture and 46.8% against CRS). The overall sensitivity of Xpert and MPCR was 16% and 72%, respectively. Xpert missed three culture-positive cases and MPCR detected additional 11. Ultra and Xpert missed two and four cases of RifR, respectively. A total of 13(59%) cases were reported 'trace' by Ultra in which RifR could not be evaluated.

Conclusion: Ultra outperformed Xpert in diagnosing OTB. The advantage of Ultra's simultaneous RifR detection is lost since the trace bacterial loads in the specimens cause indeterminate results of RifR testing.

Abbreviations: OTB: Ocular tuberculosis; Ultra: Xpert MTB/RIF Ultra; Xpert: Xpert MTB/RIF, MPCR: multiplex polymerase chain reaction; NAATs: Nucleic acid amplification tests; MLAMP: multitargeted loop-mediated isothermal amplification; PPV: positive predictive value; NPV: negative predictive value; EPTB: extrapulmonary tuberculosis; VF: vitreous fluid; DNA: deoxyribonucleic acid; ATT: antitubercular therapy; RifR: Rifampicin resistance; RifS: Rifampicin susceptible; RifI: Rifampicin indeterminate.

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IS1081; IS6110; MDR-TB; molecular diagnosis; tuberculous uveitis

Ocular tuberculosis (OTB) is an important form of extrapulmonary tuberculosis (EPTB). The incidence of OTB has increased over the last decade, both in the countries endemic for tuberculosis like India[1] and even in non-endemic countries[2–4] It has substantial morbidity in the form of transient to permanent visual loss in up to 75% patients within the first six months of onset of symptoms.[5] An intriguing feature of OTB is its dual mode of acquisition. The eyes can be inflicted not only by hematogenous seeding or contiguous spread of the tubercle bacilli directly, but also as a hypersensitivity ocular response to the tubercle bacilli elsewhere in the body.[6] Irrespective of the mode of acquisition, establishing a diagnosis of OTB remains a challenge. While the paucibacillary nature of OTB hampers detection of tubercle bacilli in the uveal tissue, generating a suspicion for hypersensitivity reaction is difficult as systemic evidence of tuberculosis elsewhere in the body is present in only 20–40% patients of OTB.[7] Ocular radiological investigations like ultrasonographic imaging and angiography offer limited help[8] and the clinical diagnosis is 'presumed' to be OTB,[9] in presence of other circumstantial evidences.

The microbiological confirmation of *M. tuberculosis* on culture remains the gold standard for diagnosing OTB. However, several inherent limitations like low bacillary load and slow rate of growth of *M. tuberculosis* on culture hampers its use in active case management. Nucleic acid amplification tests (NAATs), with their speed and accuracy, have transformed the diagnosis of tuberculosis including OTB. Several NAATs, including polymerase chain reaction (PCR),[10] multiplex PCR[11] and loop-mediated isothermal amplification[12] have been developed for OTB diagnosis. Commercial NAATs like Xpert MTB/RIF assay (Xpert)[13] and MTBDRplus assay[14] offer the added advantage of simultaneous detection of drug resistance, a phenomenon not uncommon in OTB.[15]

Xpert MTB/RIF Ultra (Ultra) is the second generation of Xpert. Though both Ultra and Xpert are semi-automated, Ultra is the updated version with a better molecular chemistry (high resolution melt curve analysis instead of real-time PCR) and incorporation of a second gene target for *M. tuberculosis* (IS1081 in addition to IS6110).[16] These

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features have enhanced the reported sensitivity of Ultra to 15.6 colony forming units (CFU)/mL for pulmonary TB,[16] thereby approaching that of culture itself. Though the WHO guidelines 2020 recommended using Ultra as the initial test for diagnosing EPTB and detection of drug resistance,[17] its diagnostic accuracy and superiority over Xpert in various EPTB specimens, including OTB, has not yet been documented. While a comparison of Ultra and Xpert in diagnosing tuberculous meningitis (TBM) has been reported,[18] to the best of our knowledge, such comparison is lacking for OTB. Even in the two recent studies on Ultra, one Cochrane systematic review[19] and the other Belgian study on 461 EPTB samples,[20] ocular samples were not included.

The present study was therefore planned to evaluate the performance of Ultra in diagnosing OTB. Additionally, its performance was compared with Xpert as well as multiplex PCR (MPCR). The MPCR, consisting of three genes, has been developed in-house in our laboratory and patented in July 2020 by Government of India (Patent no. 340788). Though this MPCR had shown a performance superior to Xpert and MTBDRplus assay in diagnosing OTB in our previous experience,[15] we wanted to investigate it with Ultra as there is a felt need for a more objective molecular test that can be available to everyone.

Materials & methods

Study setting: Ethical approval was obtained from the Institute Ethical Committee. 75 vitreous fluid samples (VF), obtained from patients undergoing pars plana vitrectomy at the Advanced Eye Centre and processed in the mycobacterial division of the department of Medical Microbiology of our tertiary care institute were included in the study. Seventy one samples were collected prospectively between 2018 and 2021 and the remaining four were archived samples collected between 2012 and 2016. These four samples were known cases of rifampicin resistance and were intentionally included to evaluate Ultra for drug resistance.

The VF samples were grouped as follows:

Group I – Confirmed OTB (*n* = 3), those who were culture-positive;

Clinically suspected OTB (*n* = 47), those who were culture-negative and presumed to be OTB on the basis of criteria described previously.[9] Since ours is a tertiary care hospital and patient presents after referral from other healthcare centers, a clear demarcation between 'probable' and 'possible' OTB, as described by Gupta et al.[21] is not always possible. The group 'clinically suspected' included both probable and possible OTB patients.

Group II – Non-OTB, control group (*n* = 25) consisting of patients undergoing pars plana vitrectomy for non-inflammatory disorders like macular hole, retinal detachment, etc.

Informed consent was obtained from each patient and a detailed clinical history on pre-formed proforma was obtained. The initiation of therapy was independent of microbiological results and patients were followed-up for a minimum of 6 months post-treatment. An attempt was also made to analyze the treatment outcomes of patients reported as drug resistant by any of the NAATs.

Sample collection: Following aseptic precautions, a 23- or 25-gauge pars plana vitrectomy was used to collect VF sample. About 2 ml of undiluted VF sample was collected under air infusion from each patient and transported in ice boxes to the department of Medical Microbiology. Post sampling, the infusion fluid was instilled and desired ophthalmological procedure was carried out.

Sample processing: All samples were handled in biosafety cabinet level II and all standard mycobacteriological precautions were followed. Each sample was coded and randomly distributed to blind the investigator of their initial grouping. Each sample was divided into four parts as follows: 0.5 mL Ultra, 0.5 mL Xpert, 0.4 mL MPCR, 0.2 ml Lowenstein Jensen culture, and rest stored at −20°C for future reference (Figure 1).

Xpert Ultra assay on VF samples: 0.5 mL of the VF sample was added to 2 mL of sample reagent and mixed vigorously. It was then allowed to stand at room

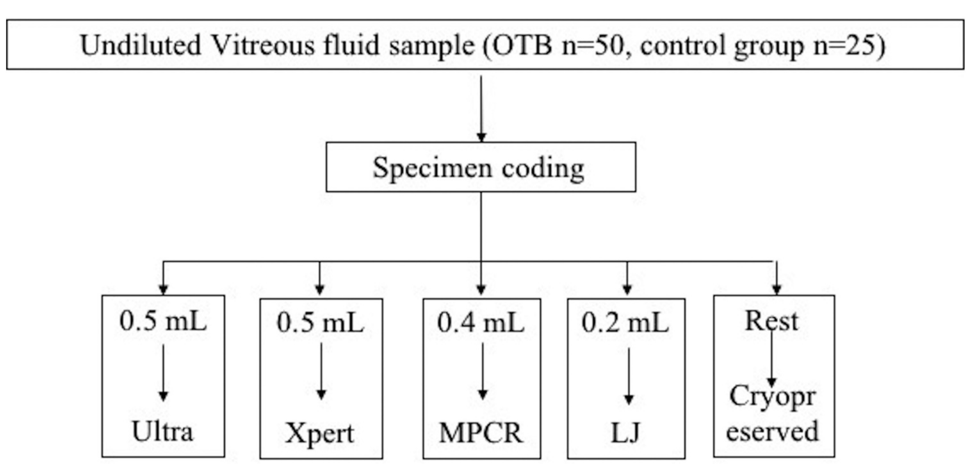

Figure 1. Diagrammatic representation of distribution of vitreous fluid specimens into different aliquots.

temperature for 15 minutes. This fluid was finally added to the Ultra cartridge and processed using GeneXpert MTB/RIF Ultra (Cepheid Inc., Sunnyvale, CA, USA), as per manufacturer's instructions. The result was reported as *M. tuberculosis* detected or not detected. For those cases wherein *M. tuberculosis* was detected, an estimate of the bacillary load was also reported as trace, very low, low, medium, and high. Rifampicin susceptibility was reported in one of the following formats – rifampicin sensitive (RifS), resistant (RifR) or indeterminate (RifI).

Xpert assay on VF samples: 0.5 mL of VF sample was added to 2 mL of sample reagent, mixed thoroughly, allowed to stand for 15 minutes and then added to the Xpert cartridge. The cartridge was then loaded into the GeneXpert MTB/RIF assay following manufacturer's instructions.

DNA extraction and MPCR: QIAamp Mini extraction kit (Qiagen, Hilden, Germany) was used to extract DNA directly from the VF samples following manufacturer's instructions. The extracted DNA was properly labelled and stored at −20°C till use. The MPCR, using three targets MPB64, IS6110 and Protein B, was performed as described previously.[15] The sample was considered positive for *M. tuberculosis* if any of the gene targets were amplified, as detected by bands at 240 bp, 123 bp, and 419 bp on gel electrophoresis.

Drug susceptibility testing (DST): Phenotypic DST (pDST) was performed for all culture-positive samples using 1% proportion method. Genotypic DST (gDST), using rpoB gene for rifampicin resistance, was performed for all samples that were reported to have *M. tuberculosis* by any of the three NAATs, Ultra, Xpert and MPCR. For all those samples reporting RifR, the *M. tuberculosis* DNA was also subjected to katG gene sequencing to determine resistance to isoniazid. The gene sequencing was performed on Big-Dye Terminator v 3.1 and ABI 3130 sequencer (Applied Biosystems, Foster City, CA).

Reference standard used for comparison: Two reference standards were used for diagnosing OTB. Culture was used as reference standard for confirmed cases and composite reference standard (CRS) [consisting of clinical features (Broad-based posterior synechiae, retinal vas-culitis with or without choroiditis, serpiginous-like choroiditis), radiological features and positive response to ATT] as described previously,[9] was used as reference standard for culture-negative cases. Rifampicin susceptibility was also evaluated using two references standards, pDST for culture-positive cases and gDST for culture-negative cases.

Statistical analysis: Standard formulae were used to calculate the sensitivity, specificity, positive predictive value and negative predictive value of the microbiological tests used.

Results

Performance of Ultra in diagnosing OTB (Table 1)

Using culture as the reference standard (n = 3), Ultra was able to detect presence of *M. tuberculosis* in all three confirmed OTB cases giving a sensitivity of 100% for culture-positive OTB cases. The bacillary load was 'very low' in all these three cases.

Table 1. Evaluation of xpert MTB/RIF ultra (Ultra), Xpert MTB/RIF (Xpert), and multiplex PCR for diagnosis of OTB.

Group	Number of cases	Culture + N (%)	Ultra N (%) +	Ultra N (%) RifS	Ultra N (%) RifR	Ultra N (%) RifI	Xpert N (%) +	Xpert N (%) RifS	Xpert N (%) RifR	Xpert N (%) RifI	MPCR + N (%)	Ultra Very Low	Ultra Trace	Xpert Very low
Confirmed OCTB cases	3	3 (100%)	3 (100%)	1 (33.3%)	2 (66.6%)	-	-	-	-	-	3 (100%)	3 (100%)	-	-
Clinically Suspected OCTB cases	47	-	22 (46.80%)	7 (31.81%)	2 (9.09%)	13 (50.09%)	8 (17.02%)	3 (37.5%)	2 (25%)	3 (37.5%)	33 (70.02%)	9 (40.90%)	13 (59.08%)	5
Total	50	3 (6%)	25 (50%)	8 (32%)	4 (16%)	13 (52%)	8 (16%)	3 (37.5%)	2 (25%)	3 (37.5%)	36 (72%)	12 (48%)	13 (52%)	5 (62%)
Control group	20	-	-	-	-	-	-	-	-	-	-			

Using CRS as the reference standard ($n = 47$), Ultra was able to detect presence of *M. tuberculosis* in 22 clinically suspected OTB cases giving a sensitivity of 46.8% for culture-negative OTB cases. The bacillary load was reported as 'very low' in 9 (40.9%) and 'trace' in 13 (59.08%) cases.

Comparative performance of Ultra, Xpert and MPCR in diagnosing OTB (Table 1, Figure 2)

The overall sensitivity of diagnosing OTB by Ultra, Xpert, and MPCR was 50% (25/50), 16% (8/50), and 72% (36/50), respectively. Among the confirmed cases, all three culture-positive cases were detected by both Ultra and MPCR but missed by Xpert.

Among the clinically suspected cases, the sensitivity of detection by Ultra, Xpert, and MPCR was 46.8% (22/47), 17% (8/47), and 70% (36/47), respectively. All cases positive by Xpert were also positive by Ultra and all cases positive by Ultra were also positive by MPCR. MPCR detected 11 additional cases of OTB that were missed by Ultra and Xpert. The sensitivity, specificity, PPV, and NPV of the NAATs is depicted in Table 2.

Performance of Ultra and Xpert in determining rifampicin susceptibility (Table 3)

Overall, 4 (16%) cases were reported as RifR by Ultra, out of which two were culture-positive and two culture-negative. Xpert reported RifR in only two cases, the same culture-negative cases as Ultra. Ultra and Xpert reported RifS in 7 and 3 cases, respectively and RifI in 13 and 3 cases, respectively. Among the cases reported RifI, Ultra reported a bacillary load of 'trace' in all 13 cases while 'very low' load was reported for the 3 RifI cases by Xpert.

rpoB gene sequencing results of cases positive by Ultra, Xpert or MPCR (Table 3)

Among the four cases reported RifR by Ultra (including two reported RifR by Xpert), three had mutations at the codon 531 and one at codon 516. On subjecting representative (21 out of 36) MPCR positive samples to rpoB gene sequencing, mutations were observed in additional two cases that were reported negative by both Ultra and Xpert. Both the samples had mutations at codon 531. Thus, Ultra missed two additional cases of RifR that were detected by MPCR.

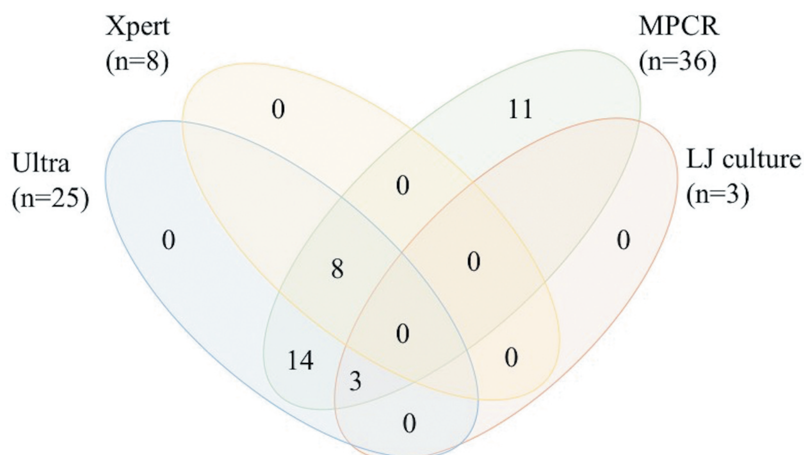

Figure 2. Venn diagram showing the number of OTB cases detected by different diagnostic tests.

Table 2. Sensitivity, specificity of Xpert Ultra, Xpert MTB/RIF, and multiplex PCR for diagnosis of OTB.

Test	Test results	GITB cases (n = 50)	Control group (n = 25)	Sensitivity % [95%CI]	Specificity % [95%CI]	PPV % [95%CI]	NPV % [95%CI]
GeneXpert	Positive	8	-25	16%	100%	100%	37.31%
MTB/RIF	Negative	42		[7.17% to 29.11%]	[86.28% to 100.00%]		[34.53% to 40.18%]
Xpert Ultra	Positive	25	-25	50%	100%	100%	50%
	Negative	25		[35.53% to 64.47%]	[86.28% to 100.00%]		[43.11% to 56.89%]
MPCR	Positive	36	-25	72%%	100%	100%	64.10%
	Negative	14		[57.51% to 83.77%]	[86.28% to 100.00%]		[53.38% to 73.58%]
Culture	Positive	3	-25	6%	100%	100%	34.72%
	Negative	47		[1.25% to 16.55%]	[86.28% to 100.00%]		[33.15% to 36.33%]

Table 3. rpoB gene sequencing results of Xpert MTB/RIF, Ultra, and MPCR positive cases.

Test (Number of positive cases)	RIF Sensitive	RIF Resistant	RIF Indeterminate	rpo B gene sequencing RIF Sensitive Number(%)	rpo B gene sequencing RIF Resistant Number(%)	Codon at which mutations were observed
Xpert MTB/RIF (8)	3 (37.5%)	2 (25%)	3 (37.5%)	6 (75%)	2 (25%)	531(1) 516(1)
Xpert Ultra (25)	8 (32%)	4 (16%)	13 (52%)	21 (84%)	4 (16%)	531(3) 516(1)
MPCR (36*)	-	-	-	15	6 (16.66%)	531(4) 526(1) 516(1)

In 36* MPCR positive cases rpoB gene sequencing was carried out in 6 resistant cases detected by other tests and 15 representative sensitive cases.

Table 4. Clinical profile, investigations, treatment, and outcome of patients reported as drug resistant by any of the tests.

							Mutated codons		Visual acuity				
Case	Age/sex	TST	Diagnosis	Ultra	GX	MPCR results	RpoB (RIF)	KatG (INH)	Initial	Final	Treatment	Follow up	Outcome
1.*	28/F	+	LE CG	RR	RR	Pos	516 (GAC→TAC)	315 (AGC→ACC)	HM	HM	CS+ MDR TB therapy	45 m	Healed
2.*	17/M	+	RE MSC	Neg	Neg	Pos	531 (TCG→TTG)	315 (AGC→ACC)	6/9	1/60	CS+ MDR TB therapy	36 m	Healed
3.*	21/F	+	RE CR	RR	RR	Pos	531 (TCG→TTG)	-	6/12	6/9	CS+ MDR TB therapy	60 m	Healed
4.*	44/M	+	LE MSC	Neg	Neg	Pos	531 (TCG→TTG)	-	6/9	6/36	CS+ MDR TB therapy	120 m	Healed
5.#	19/M	+	LE SVV	RR	Neg	Pos	531 (TCG→TTG)	315 (AGC→ACC)	CF	CF	CS+ MDR TB therapy	6 m	LFU
6.#	45/F	+	LE Panuveitis	RR	Neg	Pos	531 (TCG→TTG)	315 (AGC→ACC)	1/60	6/36	CS+ MDR TB therapy	20 m	Healed

*Cases from previous study.[15] #Cases from present study.

TST, tuberculin skin test; CXR, chest Xray; MPCR, multiplex PCR; Rif, rifampicin; Inh, isoniazid; GX, GeneXpert; LPA, line probe assay; LE, left eye; RE, right eye; BE, both eyes; CG, choroidal granuloma; CR, choroiditis; MSC, multifocal serpiginoid choroiditis; SVV, small vessel vasculitis; Pos, positive; Neg, negative; RR, rifampicin resistant; CS, corticosteroid; MDR TB, multidrug resistant tuberculosis; ATT, anti-tubercular treatment; m, months; LFU, left to follow-up.

Analysis of the clinical profile, reporting of drug resistance by different tests and treatment outcomes of patients reported as RifR (Table 4)

Overall six cases of RifR were reported in the present study. Out of these six, *M. tuberculosis* was detected in all six by MPCR while two were missed by Ultra and four were missed by Xpert. Among the four archived samples, RifR was detected by both Ultra and Xpert in two samples (Case no. 1 and 3) while they both missed detecting *M. tuberculosis* in the other two samples (Case no. 2 and 4). Between the two new samples (Case no. 5 and 6), Ultra detected RifR in both while Xpert did not even detect *M. tuberculosis* in them. On subjecting the six RifR cases to katG sequencing, InhR was noted in four samples (case no. 1, 2, 5 and 6), labelling them multidrug resistant (MDR). Thus, Ultra and Xpert failed to detect presence of *M. tuberculosis* in one and three cases of MDR-OTB, respectively. The patients received treatment for MDR-TB along with corticosteroids and were followed up for a period ranging from 6 to 120 months. The overall outcome of RifR/MDR-OTB was favorable as OTB healed in five patients while one was lost to follow-up beyond six months.

Discussion

OTB has been a clinical and diagnostic challenge. Literature reports that OTB constitutes 10–20% of all cases of uveitis in endemic countries[22]; however, these figures are largely understated due to vague clinical features and poor yield on culture. The emerging challenge of drug-resistant OTB further compromises the visual outcome.[15] In this widening gap between diagnostic techniques and emerging resistance, the WHO 2020 recommendation of using Ultra and Xpert as the first line tests for diagnosing EPTB and rifampicin resistance was a ray of hope for OTB as well.[17] The current study, for the first time, evaluated Ultra on VF samples for the diagnosis of OTB and compared it with Xpert and MPCR.

The sensitivity of Ultra in diagnosing OTB, in the current study, was 100% against culture and 46.8% against CRS as the reference standard. The overall sensitivity in diagnosing OTB was 50%. Since the evaluation of VF samples using Ultra was carried out for the first time in the world, there is no literature available for direct comparison. A comparison can be made, however, with another EPTB sample that has been evaluated by Ultra, the cerebrospinal fluid for diagnosing TBM. The summarized reported sensitivity of Ultra in diagnosing TBM ranges from 80% to 100% against culture as reference standard and from 12.5% to 90% against CRS as reference standard.[18] Our findings of sensitivity of Ultra against culture and CRS are in tune with those of TBM. However, the overall sensitivity of Ultra in detecting OTB was lower than that reported for TBM (50% in current study vs 58–76%). This could be attributed to the varied forms of OTB, ranging from multifocal serpiginous-like choroiditis (MSC) to choroidal tubercle, that could have contributed to varied yields of tubercle bacilli in our patients. The latest British Thoracic Society clinical statement for diagnosis and management of OTB 2021 also acknowledged that PCR-positivity varied with clinical phenotype, being 40% for MSC and other choroiditis to as high as 80% for military forms and choroidal tubercles.[23]

In a head-to-head comparison between Ultra and Xpert for diagnosing OTB, the sensitivity of Ultra was superior to that of Xpert not only against culture (100% vs 0%) and CRS (46.8% vs 17%), but in overall diagnosis of OTB (50% vs 16%) as well. Similar observation was made earlier by an Indian study where Ultra was found superior to Xpert in diagnosing TBM.[18] The better yield by Ultra, in comparison to Xpert, can be attributed to incorporation of two gene targets in Ultra instead of one and replacement of simple real-time amplification chemistry of Xpert with high resolution melt-curve analysis.[16] These improvements further increase the sensitivity of detection of Ultra. For this reason, a new category called 'trace' was introduced in Ultra that signifies those cases where the bacillary low is too low. It is pertinent to note that Ultra detected 'trace' load of *M. tuberculosis* in 13 samples of OTB in the current study, which were missed by Xpert. The Global Laboratory Initiative (GLI) guidelines of the Stop-TB partnership recommend treating all cases of EPTB reported 'trace' by Ultra as true-positives and include it for clinical decision-making.[24]

RifR was reported by Ultra in four cases of OTB in the current study, two culture-positive and two culture-negative. There were no cases of false-RifR reported by Ultra as these four cases were confirmed for RifR by phenotypic and genotypic methods of DST. Similar observation was made earlier in TBM where no false-RifR was reported by Ultra.[18] Xpert reported only two cases of RifR in the present study. Though no false-RifR was reported by Xpert in the present study, Xpert has reported cases of false-RifR earlier ranging from <1% in

TBM[18] to 8% in EPTB.[19] The cases reported RifS and RifI by Ultra were also found to be susceptible to rifampicin. Hence, Ultra didn't report any false-RifS.

There were 13 cases, constituting 52% of the total cases of OTB and 59% of clinically suspected OTB, wherein Ultra reported indeterminate susceptibility to Rif. All these 13 cases had a bacillary load of 'trace.' Previous researchers have also reported similar observation of Ultra producing RifI result in cases with 'trace' load, though their number of cases were less ranging from 20%[18] to 43%[25] in TBM patients. A higher percentage of RifI in the present study could be attributed to lower bacillary load as a result of lower volume of sample collected (2 ml VF vs 3–4 ml CSF), varied clinical forms of OTB varying in their bacillary load and collection of sample after initiation of ATT. On subjecting these RifI cases to rpoB gene sequencing, as suggested by the GLI guidelines,[24] no mutations were found and they were also labelled RifS.

MPCR showed superior performance than both Ultra and Xpert in diagnosing OTB in the present study. MPCR detected 11 additional cases out of which one was RifR and another one was MRD-OTB, as confirmed by sequencing later. Similar observation was made earlier wherein MPCR outperformed both Ultra and Xpert in TBM diagnosis[18] and MPCR outperformed both Xpert and MTBDRplus assay in diagnosing OTB.[15] The better detection yield by MPCR could be attributed to the incorporation of three gene targets that enable detection of those strains which lack any of the genes. The higher yield by MPCR could also be secondary to utilization of undiluted VF specimen for DNA extraction while the sample reagent added in case of Ultra and Xpert could have diluted the specimen.

Though both Ultra and Xpert offer the advantage of simultaneous detection of RifR, they are limited in their throughput and accessibility in an endemic low-resource country like India. Further, with more than half the cases of EPTB being 'trace' category (59% in the current study), the advantage of simultaneous detection of RifR does not apply as they would be reported as RifI and would need confirmation by alternative methods. Also, with a NPV of 50%, Ultra cannot serve as a reliable test for ruling-out OTB. Thus, even though Ultra offers better detection yield than Xpert, it may be used only as one of the first diagnostic test for OTB, but definitely not the last.

The study has certain limitations. These results are based on 2 mL VF samples and cannot be extrapolated to 0.1 mL aqueous humor samples for the diagnosis of OTB. The ocular signs could result from immune response to dead bacilli as well and tests targeting DNA won't be able to differentiate between live and dead bacilli. For such differentiation, mRNA detection may be used, as described recently.[26]

It can be concluded that Ultra and Xpert offer the simultaneous detection of OTB as well as RifR and can be used as initial tests for the diagnosis of OTB. When available, Ultra should be preferred over Xpert owing to its better detection yield. However, cases reported 'trace' by Ultra require confirmation of resistance by alternative methods. MPCR along with gene sequencing offers dual advantage of increased yield of detection of OTB cases along with foolproof detection of gene mutations.

Author's contribution

Study design – KS, RS, AG; Collection of data – RS, VG, NA, MD, AG; Data analysis – MS, KS, RS; Drafting of manuscript – MS; Critical appraisal of manuscript – KS, RS, AS, AG; Study supervision – KS, RS, AG.

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